Ishan Shah

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Education

University of Waterloo **Sept. 2025 – Apr. 2030**
Bachelor of Computer Science

Experience

Axibo **Incoming**
Reinforcement Learning Intern *Cambridge, ON*

- Developing and deploying Vision-Language-Action (VLA) models and Reinforcement Learning (RL) frameworks to enhance autonomous bimanual manipulation and locomotion capabilities for humanoid robotic platforms.

WAT.ai **Jan. 2026 - Present**
Machine Learning Researcher (Robotics) *Waterloo, ON*

- Researched **preference learning** and **human-feedback-driven optimization** for sequential decision-making models.
- Designed and implemented **realistic robotic simulation environments** and trained **Actor-Critic (ACT) policies**, evaluating robustness across diverse scenarios.
- Built and optimized **Python/PyTorch learning pipelines** and scalable human-in-the-loop setups for improving policy performance under sparse or under-specified rewards.

IPMD **July 2025 - Sept. 2025**
Software Engineer Intern *San Mateo, CA*

- Replaced custom auth logic with **Flask-JWT-Extended**, centralizing token management across **10 endpoints** and resolving registration edge cases.
- Modularized frontend by extracting **5+ reusable components**, reducing code duplication by **30%** and accelerating future feature development.
- Built a real-time **computer vision–driven** emotion analysis pipeline integrated into a React (TypeScript) chat interface, surfacing live sentiment feedback during conversations.

Technical Skills

Languages: Python, C++
Frameworks: PyTorch, JAX, ROS2, OpenCV, NumPy
Simulation: MuJoCo, Isaac Lab, Gymnasium
Tools/Deployment: Linux, Git, Docker

Projects

Human-in-the-Loop Teleoperation for Imitation Learning | *MuJoCo, MediaPipe, Pytorch*

- Designed a **low-latency teleoperation** system to control a simulated **ALOHA bimanual robot** using monocular webcam-based hand tracking, eliminating specialized hardware.
- Built a real-time human-to-robot control pipeline using **Inverse Kinematics** to map tracked hand poses into robot joint-space commands, refining **quaternion-based** mappings to avoid gimbal lock.
- Developing a scalable **HDF5-based** demonstration logging pipeline structured for training **Action-Conditioned Transformers (ACT)** and **Diffusion Policy** models.

Reinforcement Learning for Continuous Control | *Python, PyTorch, Gymnasium, Numpy*

- Developed a from-scratch implementation of **Proximal Policy Optimization (PPO)** with **Generalized Advantage Estimation (GAE)** to solve continuous action space environments.
- Architected a custom **Actor-Critic** network in **PyTorch**, optimizing hyperparameters to achieve convergence on the BipedalWalker-v3 environment.
- Implemented **orthogonal weight initialization** and **learning rate annealing**, improving training stability and reducing variance in episodic returns.